Muhammad Yahya Azeem

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Professional Summary

Software and security researcher specializing in systems programming, emulation layers, machine learning optimization, and offensive security.

Technical Skills

Programming Languages: Rust, C, C++, Python, TypeScript, Shell, Assembly.
Systems & Architecture: OS Kernel Development, Virtual Memory Management, Vulkan SDK, Systems Virtualization.
Deep Learning & Optimization: Transformer Architecture Design, Custom Optimizers, Quantization, PyTorch, NumPy.
Offensive Security: Exploit Development, Kernel-level Exploitation, Reverse Engineering, Binary Analysis, Network Auditing.

Professional Experience

Cybervigilance LLC — Technology Risk Management & Cybersecurity Intern *Oct 2025 – Present*

- **GRC & Risk Management:** Delivered 3 risk assessments and corporate presentations aligned with NIST frameworks and ISO 27001 standards for multi-million and billion-dollar entities.

Cybersecurity & Offensive Capabilities

Hack The Box (HTB) — Rooted "Cobblestone" Machine (Insane Difficulty) *June 2026*

- **Active Exploitation:** Compromised and rooted the machine in 3 days utilizing binary analysis and custom payload bypasses.

Projects

ShadPS5 — PlayStation 5 Emulator *June 2026*

Key Technologies: Rust, Vulkan 1.3, SDL2, ash

- **ELF/SELF Loader & Linker:** Designed the loader in Rust, translating 200+ Cryptographic Name Identifiers (NIDs) to initialize guest library imports.
- **GPU Shader Translation:** Engineered a translation engine mapping 50+ guest shader instructions to host-native Vulkan SPIR-V bytecode to render 3D geometry.

Project Golf Ball — OpenAI Parameter Golf LLM Optimizer *April 2026*

Key Technologies: Python, PyTorch, CUDA

- **11-Layer Transformer Model:** Optimized an 11-layer language model (512-dim) trained on 100M+ FineWeb tokens to fit strictly within a 16MB artifact limit.

Polymarket Copytrade Bot *May 2026*

Key Technologies: Rust, Tokio, WebSockets, CLOB

- **Asynchronous Copytrading Engine:** Developed a Rust/Tokio system to mirror Whale orders with sub-5ms latency, protecting capital with a 20% daily drawdown threshold.

Other Projects: **Unixy-System** (custom OS kernel in C/Assembly supporting x86/MIPS/PowerPC) · **Protection Valley** (SvelteKit 5 e-commerce platform with a Vercel-deployed Rust serverless backend).

Education

The University of Texas at Dallas *2026 – 2030*
Bachelor of Science in Computer Science *Richardson, Texas*